

Sharon Kodali

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Education

University of California – San Diego

B.S. Mathematics–Computer Science
Minor in Cognitive Science (Machine Learning)

Graduation 2029

GPA: 4.0

Professional Experience

UCSD Institute of Engineering in Medicine

Lead Research Intern

San Diego, CA

2024 – 2025

- Led a 5-person team analyzing 500+ DRG neuron microscopy images using Python, MATLAB, and ImageJ to extract morphological and intensity features.
- Authored and presented technical analyses contributing to 2 peer-reviewed abstracts at the Biomedical Engineering Society (BMES).

San Diego Supercomputer Center (SDSC)

Research Intern

San Diego, CA

Jun 2024 – Aug 2024

- Built an AI training catalog using Python, Jupyter, and Linux HPC systems improving ML workflow accessibility for 100+ researchers.
- Performed benchmarking and runtime analysis across supercomputing clusters to optimize large-scale ML experiments.

Open Coding Society

Project Manager and Developer

San Diego, CA

Nov 2024 – Jun 2025

- Led Agile development with Git/GitHub workflows coordinating requirements across multiple campuses.
- Built a full-stack task management and planning system using Spring Boot, SQL, REST APIs, and frontend development.

Mathnasium

Lead Math Instructor

San Diego, CA

2023 – 2025

- Taught K-12 math from Pre-Algebra to Calculus in sessions of 4–5 students using individualized instruction.
- Mentored a 10+ instructor team and supported onboarding and scheduling operations.

Projects

Spotify Visual Art Generator

Jan 2026 – Present

- Built a Python ML pipeline using NumPy, pandas, and scikit-learn analyzing 1,000+ Spotify audio feature vectors to classify playlist vibe and generate visual art with Seaborn and Matplotlib.
- Integrated the Spotify Web API with Flask REST endpoints and JavaScript frontend to generate real-time visualizations from playlists.

California Real Estate Intelligence Tracker

Oct 2025 – Nov 2025

- Developed a full-stack Python/Flask platform aggregating 5,000+ Zillow listings using Google Maps API.
- Built an indexed SQL database enabling interactive map-based property filtering by price, ROI, and location radius.

Technical Skills

Programming: Python, Java, SQL, C

Machine Learning: NumPy, pandas, scikit-learn

Visualization: Plotly, Tableau

Backend: Spring Boot, REST APIs, MySQL, PostgreSQL

Tools: Git, Linux, Jupyter Notebooks